DATA ANALYTICS PROJECT

Amazon Sales Data Cleaning, Analysis & Insights

Exploratory analysis of product pricing, discounts, reviews, brand performance, sponsored listings, and coupon behavior.

Created by Rahma Ashraf Ali Faragallah

Matplotlib • Seaborn • Pandas

2026

Executive Summary

The analysis moves from data quality to commercial patterns and actionable findings.

ORIGINAL DATASET

42,675 rows

Before duplicate removal

DUPLICATES REMOVED

962

Fully duplicated rows

ORIGINAL COLUMNS

16

All initially stored as text

KEY SALES SIGNAL

r = 0.30

Reviews ↔ estimated monthly sales

INSIGHT

The market is highly price-skewed

Most products cluster at relatively low prices, while a small number of expensive products create a long right tail. Median-based summaries are therefore more representative than the mean.

INSIGHT

Promotions are not automatically performance drivers

Discount percentage shows almost no linear relationship with review count (r = 0.02) and only a weak relationship with estimated sales (r = 0.04). Coupons are associated with lower average estimated sales in this dataset, so correlation should not be interpreted as causation.

Data Quality & Cleaning

Cleaning decisions were designed to preserve information while making the fields analytically usable.

01

Duplicate removal

Removed 962 fully duplicated rows.

02

Numeric conversion

Converted ratings, reviews, prices and sales estimates from text to numeric fields.

03

Missing values

Rating missing values were intentionally preserved as NaN rather than imputed.

04

Price logic

Interpreted “No Discount” as listed price = current price, giving 0% discount.

05

Badge parsing

Split mixed badge text into separate boolean indicators such as Best Seller and Sponsored.

06

Feature engineering

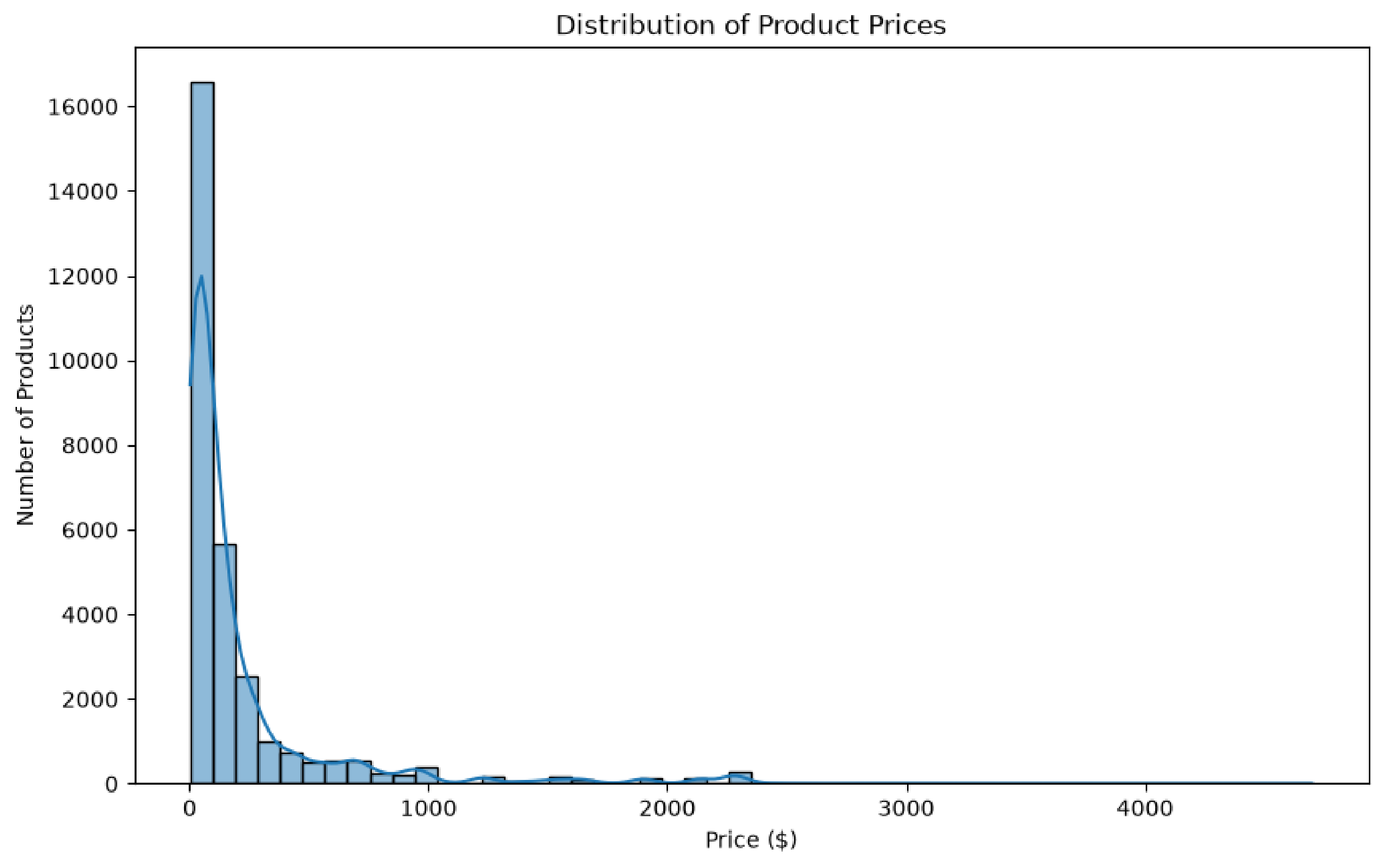
Created discount_percentage, bought_last_month_num, brand and other analysis-ready features.

INSIGHT

Why it matters

1. Product Price Distribution

Histogram of current/discounted product prices.



INSIGHT

Strong right skew

The distribution is concentrated at lower price points, followed by a long tail of expensive products.

INSIGHT

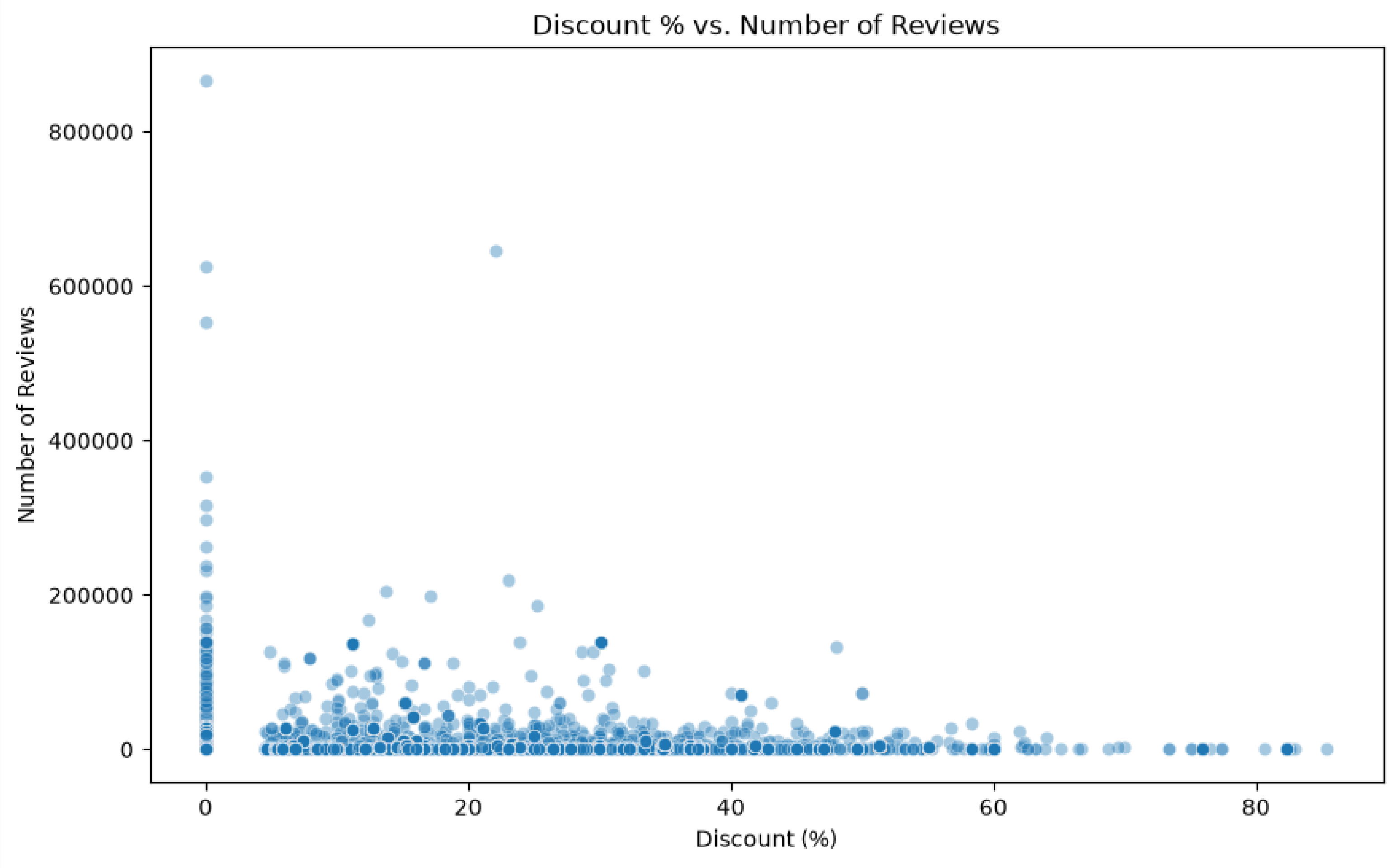
Use the median for “typical” price

A small number of high-priced products can pull the arithmetic mean upward and make it less representative.

Business implication: the dataset is dominated by accessible price points, with a smaller premium segment.

2. Discount % vs. Number of Reviews

Testing whether larger discounts are associated with stronger customer engagement.



INSIGHT

Very weak relationship

The correlation between discount percentage and review count is only 0.02, indicating essentially no linear association.

INSIGHT

Reviews concentrate at low discounts

The highest review-count outliers appear mostly around low discount levels rather than at extreme discounts.

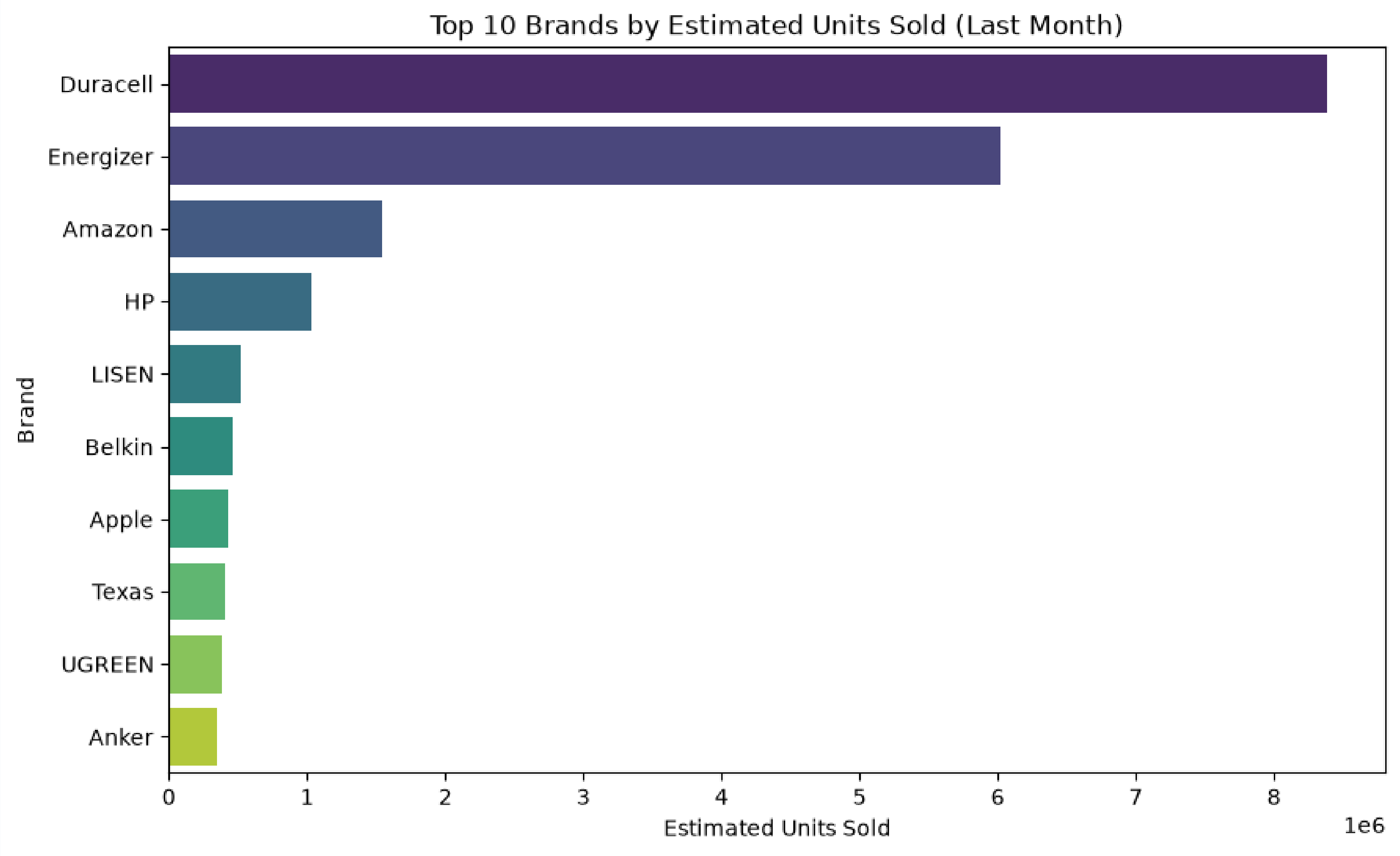
INSIGHT

Takeaway

Discount depth alone does not explain customer engagement.

3. Top 10 Brands by Estimated Units Sold

Estimated units sold in the last month, aggregated by brand.



INSIGHT

Clear brand concentration

Duracell and Energizer are far ahead of the remaining brands in estimated monthly unit sales.

INSIGHT

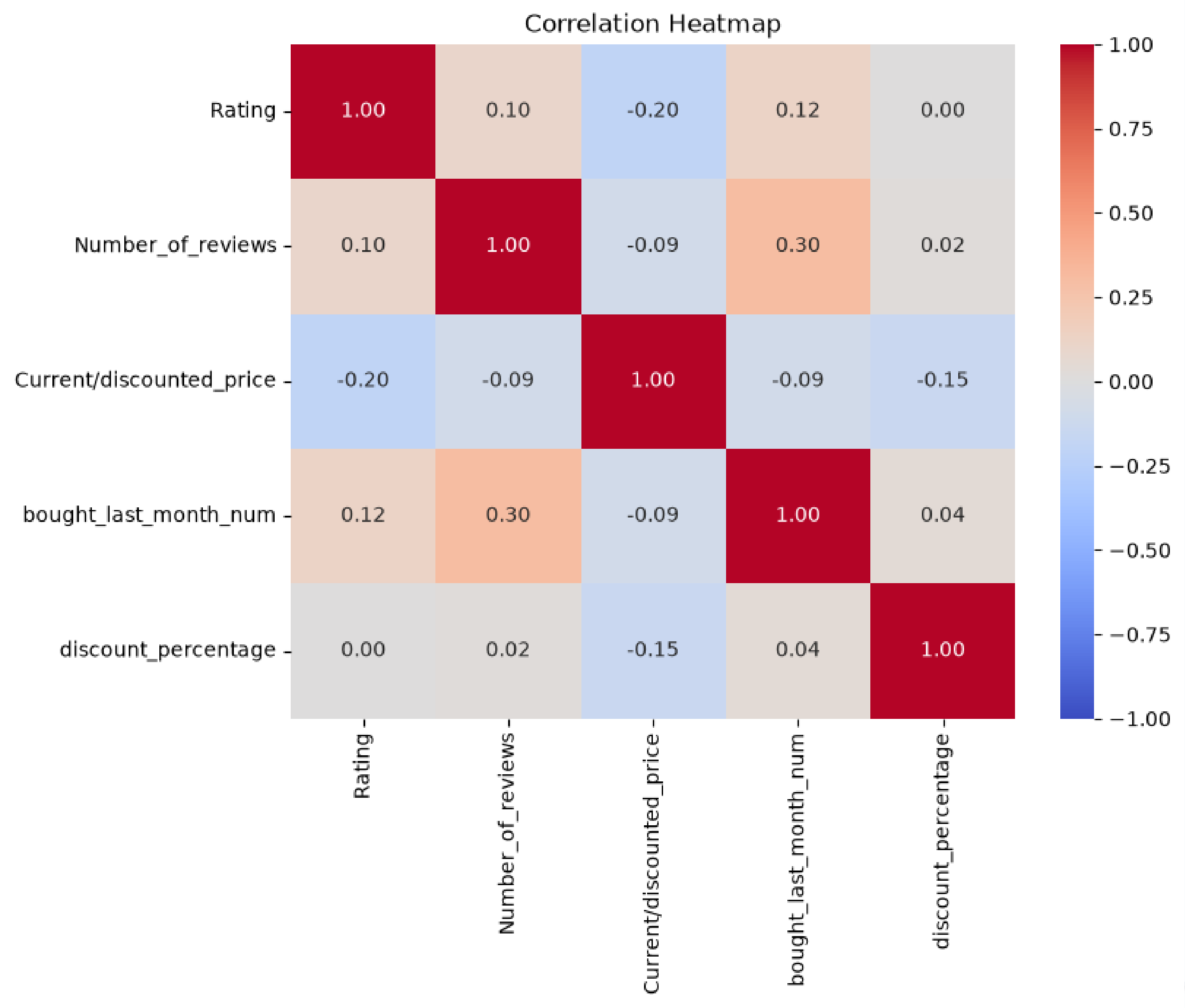
Long competitive tail

After the two leading brands, sales fall sharply, suggesting a much more fragmented second tier.

Management question: what product, pricing, or visibility factors explain the leaders' scale?

4. Correlation Analysis

Linear relationships among rating, reviews, price, estimated sales and discount percentage.



REVIEWS ↔ SALES

0.30

Weak-to-moderate positive

RATING ↔ SALES

0.12

Very weak positive

PRICE ↔ SALES

-0.09

Negligible negative

DISCOUNT ↔ SALES

0.04

Negligible

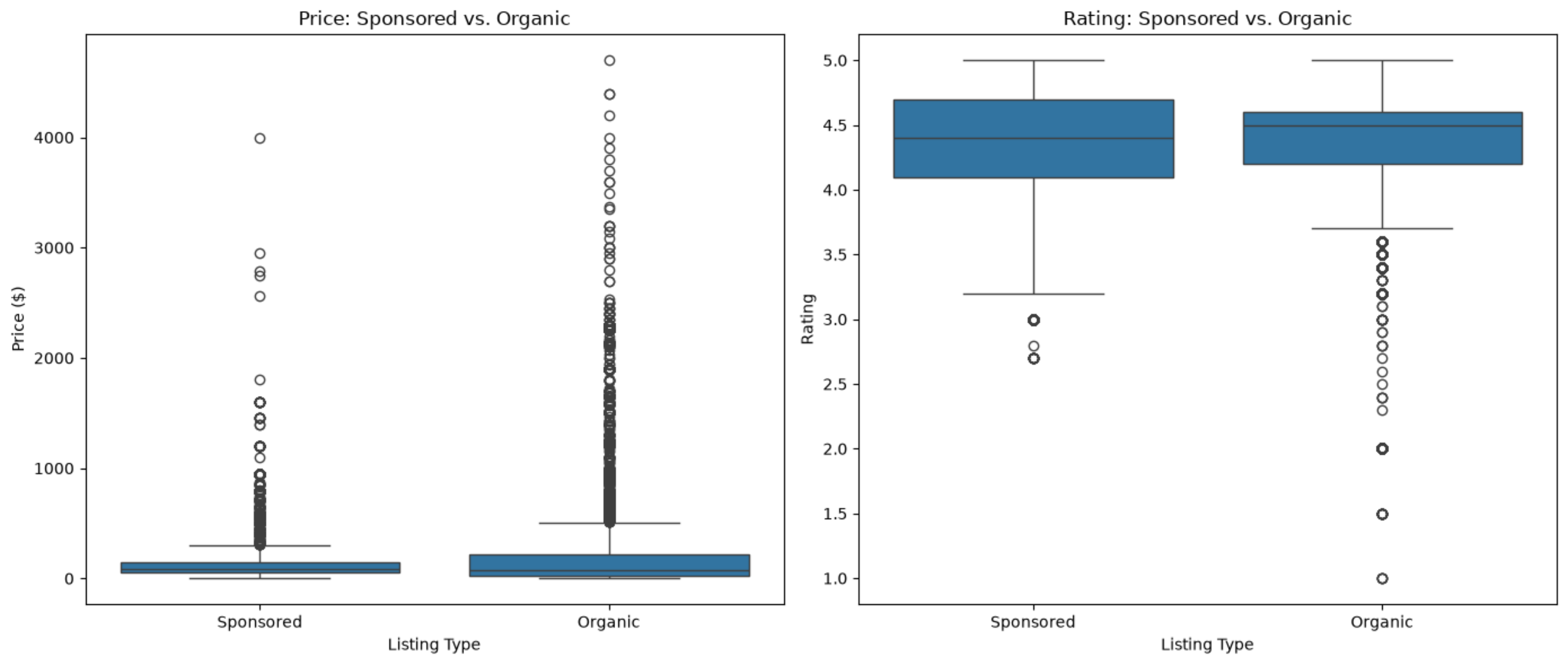
INSIGHT

Most useful signal: review volume

Among the tested variables, review count has the strongest positive association with estimated sales—but 0.30 is still far from a strong predictive relationship.

5. Sponsored vs. Organic Listings

Comparing price and rating distributions by listing type.



INSIGHT

Price distributions overlap heavily

Organic listings show a slightly higher central price, while both groups contain many high-price outliers.

INSIGHT

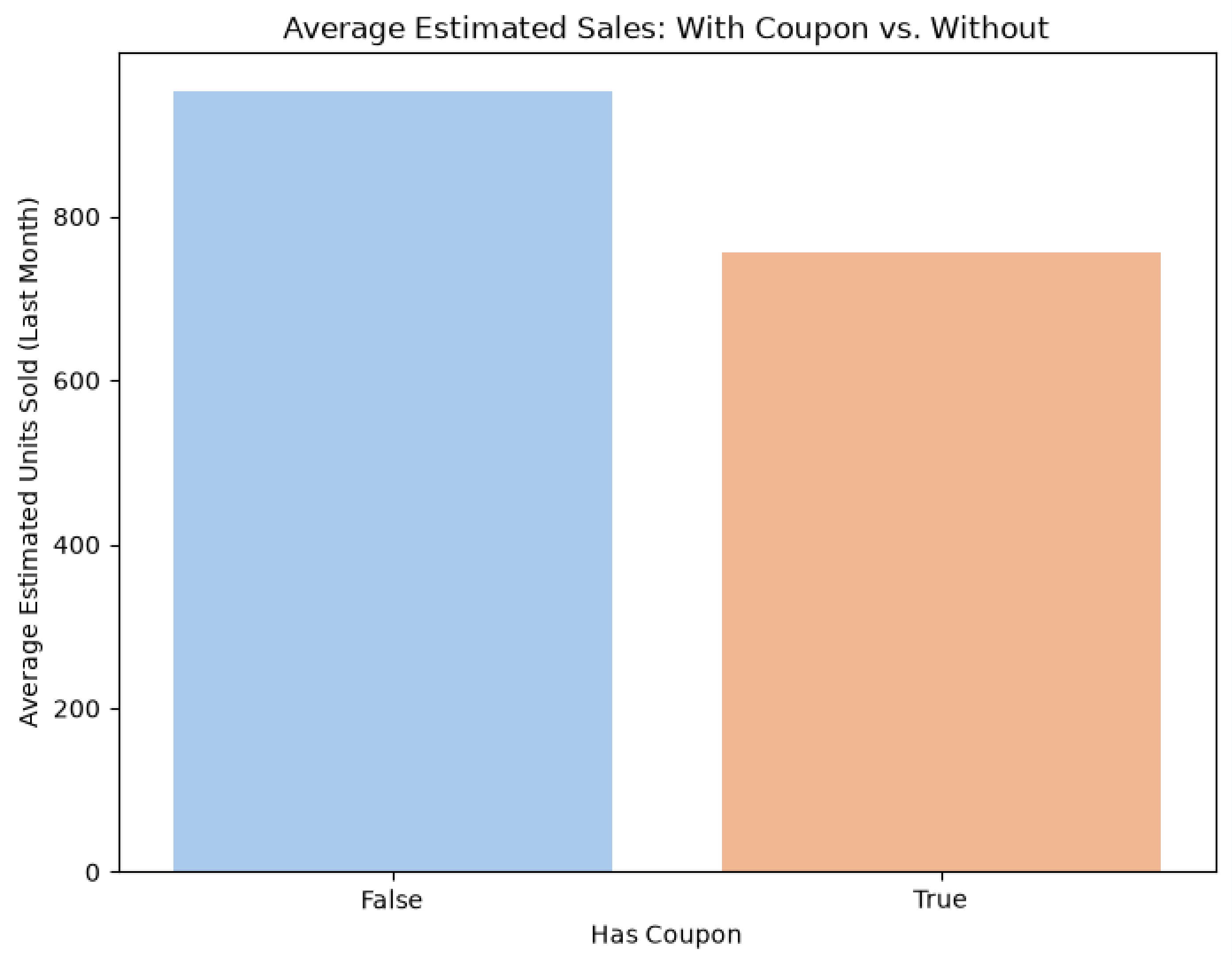
Ratings are broadly similar

The medians are close, so sponsored placement should not be treated as a proxy for higher or lower product quality.

Interpretation: advertising placement and product quality appear to be largely separate dimensions in this sample.

6. Coupon Effect on Estimated Sales

Average estimated units sold for products with and without coupons.



INSIGHT

Lower average sales with coupons

Products without coupons show a higher average estimated monthly sales volume than products with coupons in this dataset.

INSIGHT

Do not infer causation

A coupon may be a response to weak demand rather than the reason for weak demand. A controlled or time-based analysis would be needed to estimate causal impact.

Next step: compare sales before/after coupon activation or control for product category, brand and price.

Integrated Business Insights

What the six visual analyses tell us together.

Price structure

01

The catalog is heavily concentrated at lower price points, with a small premium tail.

Brand power

02

Estimated unit sales are strongly concentrated in a few leading brands—especially Duracell and Energizer.

Customer engagement

03

Review volume has the strongest tested association with estimated sales, but the relationship remains moderate.

Discounting

04

Discount percentage is not strongly related to reviews or estimated sales in this cross-sectional dataset.

Advertising

05

Sponsored and organic listings have broadly similar rating distributions; sponsorship is not a quality signal.

Coupons

06

Coupon presence is associated with lower average estimated sales, suggesting coupons may often target slower-moving products.

Recommendations & Next Analytical Steps

Turning descriptive findings into stronger business decisions.

1	**Segment the catalog**	Analyze price bands, product categories and brands separately to avoid mixing very different markets.
2	**Investigate leaders**	Deep-dive into Duracell and Energizer to identify the drivers behind their estimated unit-sales advantage.
3	**Measure promotions properly**	Use before/after comparisons or matched products to evaluate discounts and coupons instead of relying on simple correlations.
4	**Add time-based analysis**	If multiple collection dates are available, track price, promotions and estimated sales over time.
5	**Improve predictive modeling**	Combine brand, price, rating, reviews, sponsored status, coupons and category features in a supervised model.
6	**Validate data assumptions**	Review brand extraction limitations and the meaning/accuracy of estimated “bought last month” values before operational use.

KEY TAKEAWAY

Clean data turns raw marketplace records into decisions—not just charts.

The strongest story in this analysis is not that one promotion “causes” sales.
It is that brand, product economics and customer engagement interact in a complex market.

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Thank you